Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2019-P1-Model)** **Date: 13-05-2026**

Time: 09.00Am to 12.00Pm **GTA-32** **Max. Marks: 186**

KEY SHEET

PHYSICS

1	**B**	2	**A**	3	**A**	4	**C**	5	**BC**	6	**AD**
7	**ABD**	8	**BC**	9	**CD**	10	**BCD**	11	**AD**	12	**ACD**
13	**4.65**	14	**16**	15	**11.18**	16	**0.75**	17	**0.22**	18	**13.65–13.67**

CHEMISTRY

19	**A**	20	**C**	21	**D**	22	**C**	23	**B**	24	**ABCD**
25	**ACD**	26	**ACD**	27	**ACD**	28	**BC**	29	**CD**	30	**ABCD**
31	**5.8 – 5.9**	32	**28**	33	**2**	34	**4**	35	**2**	36	**3**

MATHEMATICS

37	**D**	38	**B**	39	**C**	40	**C**	41	**BCD**	42	**ABC**
43	**BD**	44	**B**	45	**ABC**	46	**ABC**	47	**CD**	48	**AC**
49	**3**	50	**1**	51	**3**	52	**6**	53	**12**	54	**8**

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SOLUTIONS

PHYSICS

1. $r=\dfrac{\left\{1+\left(\dfrac{dy}{dx}\right)^2\right\}^{3/2}}{\dfrac{d^2y}{dx^2}}$ and $a=\dfrac{V^2}{r}$

2. The size of the cube is $L\propto m^{1/3}$, so $A\propto m^{2/3}$ since the density is constant. Constant speed requires $m\propto P=Fv\propto Av^3\propto m^{2/3}v^3$.

3. Use the energy conservation principle, angular momentum conservation about edge of the step and also use the condition normal contact force at the edge of the step is greater than or equal to zero.

4. $\dfrac{1}{2}mv_0^2-\dfrac{GMm}{R}=\dfrac{1}{2}mv^2-\dfrac{3}{2}\dfrac{GMm}{R}$

   After collision $\dfrac{1}{4}mv^2-\dfrac{3GMm}{R}=0 \qquad \therefore V_0=\sqrt{\dfrac{11GM}{R}}$

5. $\dfrac{VT}{V-20}=18 \qquad \dfrac{VT}{V+20}=6$

   Solving V = 40 km/hr T = 9 min

6. $f=\dfrac{mr\alpha}{2} \qquad f=\mu mg$

   $\alpha=\dfrac{2\mu g}{r} \qquad \alpha_{cm}=\mu g$

   Displacement of the bottom most point $x=v_0t-\dfrac{1}{2}\mu gt^2-\dfrac{1}{2}r\alpha t^2$

   Hence work done by friction = $W_f=-(\mu mgx)=-\mu mg\left(v_0t-\dfrac{3}{2}\mu gt^2\right)$

   Once pure rolling at time $2t_0$ is equal to work done till $t_0 W_f=-\mu mg\left(v_0t_0-\dfrac{3}{2}\mu gt_0^2\right)$

   Work done by friction while pure rolling is zero.

7. $V=ax^{4/3}$

   $\rho=-\varepsilon_0\Delta^2V=-\varepsilon_0\dfrac{d}{dx}\left\{\dfrac{d}{dx}\left(ax^{4/3}\right)\right\} \qquad =-\dfrac{4}{9}a\varepsilon_0x^{-2/3}$

   $j=neV=\rho V$

   Calculation of V;

   $\Delta U+\Delta K=0\Rightarrow\dfrac{1}{2}mV^2=eV \qquad \Rightarrow V(x)=\left\{\dfrac{2e}{m}ax^{4/3}\right\}^{1/2}$

   $\therefore j=\dfrac{4}{9}a\varepsilon_0x^{-2/3}\sqrt{\dfrac{2ea}{m}}x^{2/3} \qquad =\dfrac{4}{9}\varepsilon_0a^{3/2}\sqrt{\dfrac{2e}{m}}$

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8. $\overline{E}' = \overline{E} + \overline{V_0} \times \overline{B} \quad \overline{B}' = \overline{B} - \dfrac{\overline{V_0} \times \overline{E}}{C^2}$

9. $B = \mu_0 ni; \Delta p = \dfrac{B^2}{2\mu_0} = \dfrac{\mu_0^2 n^2 i^2}{2\mu_0} = \dfrac{\mu_0 n^2 i^2}{2}$

$\Delta p \cdot \cancel{\ell} \cdot R \cancel{d\theta} = T \cancel{d\theta} \cdot n \cancel{\ell}$

$\Rightarrow T = \dfrac{\Delta P \cdot R}{n} = \dfrac{\mu_0 n i^2 R}{2} < F_\ell \qquad \Rightarrow i \le \sqrt{\dfrac{2F_\ell}{\mu_0 nR}}$

10. $i = \dfrac{Bav}{R} \qquad \dfrac{mvdv}{dx} = F = -\dfrac{B^2 d^2 v}{R}$

$mv_0 = \dfrac{B^2 a^2 \ell}{R} \qquad \ell = \dfrac{Rv_0 m}{B^2 a^2} \le a \qquad v_0 \le \dfrac{B^2 a^3}{mR}$

11. We could first carefully warm up the super cooled water from $-10^\circ C$ to $0^\circ C$, and then freeze the water at its normal freezing point. In the first part of this process, the heat that has to be supplied is 42 kJ, during the freezing stage the heat released is 334 kJ.

$-Q + 21kJ = 42kJ - 334kJ$

From this, $Q = 313kJ$

12. $\Delta P = -B\dfrac{\partial S}{\partial x}$

13. $Q = (M_N + M_n) - (M_B + M_{He})$ $= -0.15 MeV$

$= -0.000165 \times 931.5$

14. $\rho = \dfrac{\ell}{2}\dfrac{\cot\theta}{\cos^2\theta}$ $= \dfrac{32}{2} \times \dfrac{3}{4} \times \dfrac{4}{3}$ $= 16$ m

15. $\left|\overline{a}'\right| = \left|\overline{a} - \overline{\omega} \times \left(\overline{\omega} \times \overline{r}\right) - 2\left(\overline{\omega} \times \overline{V}'\right)\right|$ $= \sqrt{10^2 + 5^2} = 5\sqrt{5}\, m/s^2$

16. $F = \dfrac{m}{2}\omega^2 \times \dfrac{3\ell}{4} = \dfrac{3}{8} m\omega^2 \ell$

17. $i = \dfrac{\varepsilon\left\{\dfrac{L_1}{R_1} - \dfrac{L_2}{R_2}\right\}}{L_1 + L_2} = \dfrac{10\left\{\dfrac{L}{2} - \dfrac{2L}{5}\right\}}{3L} = 0.33$

18. $i = 10\sqrt{2}\left\{\dfrac{1}{5 + 5\sqrt{3}j} + \dfrac{1}{5\sqrt{3} - 5j}\right\} = \dfrac{10\sqrt{2}}{5}\left\{\dfrac{1 - \sqrt{3}\hat{j}}{4} + \dfrac{\sqrt{3} + j}{4}\right\}$

$= \dfrac{1}{\sqrt{2}}\left\{\left(1 + \sqrt{3}\right) + \left(1 - \sqrt{3}\right)j\right\}$

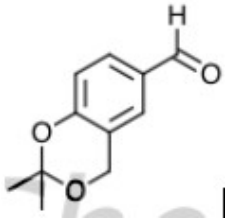
$P = V_{rms}\, i_{rms} \cos\phi$

$= 10 \times \dfrac{1}{2} \times \left(1 + \sqrt{3}\right) = 5\left(1 + \sqrt{3}\right)$

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CHEMISTRY

19. Formylation followed by formation of acetal.



20. Number of moles:n = $5\times10^{-7}\,mol$

Number of molecules:N = 3×10^{17}

Area per molecule (hexagonal packing): $A_m = 2.1625\times10^{-15}\,cm^2$

Total surface area: $A = 6.49\times10^{2}\,cm^2$

21. NCERT lab manual.

22. All are correct statements

23. Enantiomers have same molar enthalpy.

24. NCERT

25. Dichromate can not be oxidized.

26. Sigma bond is formed from $2p_z$ orbital and pi bond from $2p_x$ and $2p_y$.

27. For 100 O^{2-} ions, total Fe ions = 93. Let $Fe^{3+} = x$ and $Fe^{2+} = (93-x)$.

Charge balance: 3x + 2(93 − x) = 200 $\Rightarrow$ x = 14.

Hence $Fe^{3+} = 14$ and $Fe^{2+} = 79$.

A is correct: $(14/93)\times100 \approx 15.05\%$.

B is incorrect: density depends on mass per unit cell, not directly on Fe fraction.

C is correct: $Fe^{2+} = 79$.

D is correct: Pt electrode potential depends on Fe^{3+}/Fe^{2+} ratio.

28. $4NaC\ell + MnO_2 + 4H_2SO_4 \longrightarrow MnC\ell_2 + 4NaHSO_4 + 2H_2O + C\ell_2$ 29. A is phthalic acid, B is phthalimide, C is N-benzyl derivative.

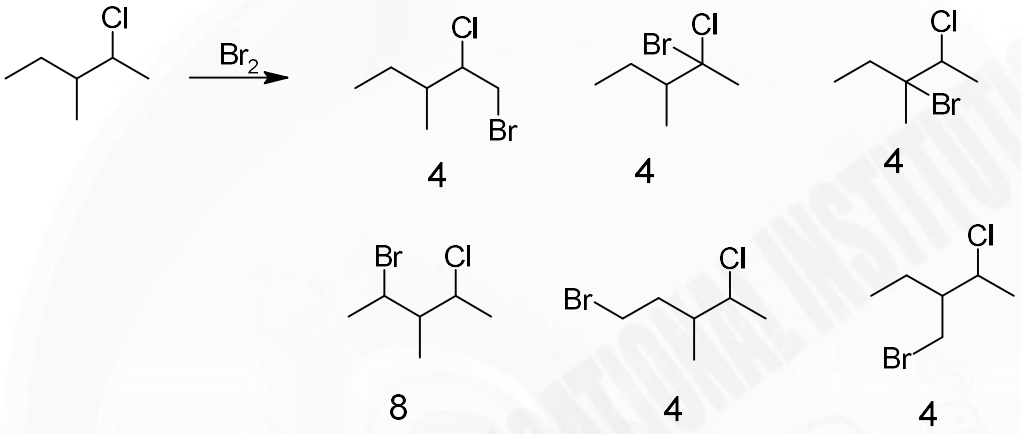
30. Periodic acid cleavage gives Formaldehyde followed by benzyl alcohol.

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31. $0 = -10000 + RT\ell n\left(\dfrac{P_{CO_2}}{P_{CO}}\right)^3$

$Q_2 = \dfrac{P_{CO}^2}{P_{CO_2}} =$

$\Delta G_2 = \Delta G_2^\circ + RT\ell nQ_2$

32. 

33. P = 6 (benzene ring has 6 sp^2 carbons). Moles forming micelles =

$(5.2-1.2)\times10^{-3} = 4.0\times10^{-3}\,mol \Rightarrow$ molecules = 2.41×10^{21}.

$Q = \left(2.41\times10^{21}\right)/\left(3.011\times10^{19}\right) = 80 \Rightarrow P\times Q = 480$

34. Each acac ligand forms one chelate ring. With 3 ligands: $N_{rings} = 3$.

Each acac ligand has 3 sp^2 carbons.

Total $N_{sp^2} = 3\times3 = 9$.

acac coordinates through oxygen atoms only → no Fe–C bond.

35. Lowering ratio = $1+\alpha$.

At 1 minute: $1+\alpha = 1.5 \Rightarrow \alpha = 0.5$, so 1 minute equals one half-life.

Required lowering = $1.75 \Rightarrow \alpha = 0.75$.

For first order: $\alpha = 1-(1/2)^n \Rightarrow 0.75 = 1-(1/2)^n \Rightarrow n = 2$.

Time = **2 minutes**.

36. Total equilibrium pressure remains 3 atm.

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MATHEMATICS

37. Equation of tangent is $y=4x\pm\sqrt{16a^2+b^2}$ this normal passes through $(-2,0)$

$\Rightarrow 0=-8\pm\sqrt{16a^2+b^2};64=16a^2+b^2$

Giving $AM\geq GM$

$$\frac{16a^2+b^2}{2}\geq\sqrt{16a^2b^2}$$

$$\frac{64}{2}\geq\sqrt{16a^2b^2} \qquad 4ab\geq 32 \qquad ab\leq 8$$

38. $y=\ell n^2x-1 \qquad y'=\frac{2\ell nx}{x}=0\Rightarrow x=1$

$x>1, y\uparrow$ and $0<x<1, y\downarrow$

$$A=\left|\int_{1/e}^{e}\left(\ell n^2x-1\right)dx\right|$$

$$=\left|x\ell n^2x\right]_{1/e}^{e}-2\int_{1/e}^{e}\left(\frac{\ell nx}{x}\right)\cdot xdx-\left(e-\frac{1}{e}\right)\Bigg|$$

$$=\left|\left(e-\frac{1}{e}\right)-2\int_{1/e}^{e}\left(\frac{\ell nx}{x}\right)\cdot xdx-\left(e-\frac{1}{e}\right)\right|$$

$$=\left|-2\left[x\ell n\right]_{1/e}^{x}-\int_{1/e}^{e}dx\right|=\left|-2\left[2\left(e+\frac{1}{e}\right)-\left(e-\frac{1}{e}\right)\right]\right| \qquad =\left|\frac{4}{e}\right|=\frac{4}{e}$$

39. Take a general point on L: $Q(1+3t,-2-2t,5+t)$

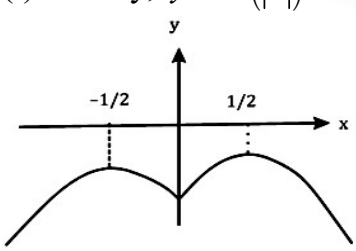
Since distance is measured parallel to plane P, $\overrightarrow{AQ}\cdot(2,1-1)=0$

$(3t-1,-2t-1,t+1)\cdot(2,1,-1)=0 \qquad \Rightarrow 3t=4\Rightarrow t=\frac{4}{3}$

Thus, $Q=\left(5,-\frac{14}{3},\frac{19}{3}\right)$ $\qquad AQ=\sqrt{(3)^2+\left(-\frac{11}{3}\right)^2+\left(\frac{7}{3}\right)^2}=\frac{\sqrt{251}}{3}$

40. $f(x)=(x-1)^3$

(i) Clearly, $y=h(|x|)$ is non–derivable at exactly one point.



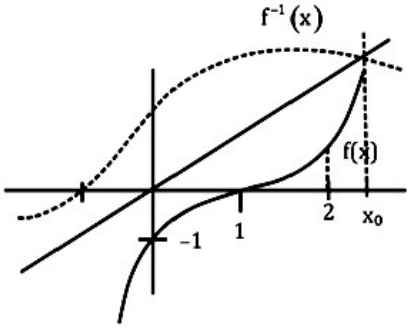
$|A|=1$

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(ii) $f(x)$ and $f^{-1}(x)$ meet on the line $y=x$ at $x=x_0$

Where $2<x_0<3$

$\{\because f(2)=1$ and $f(3)=8\}$



$$\therefore \cos^{-1}(\cos 2x_{00})+4\tan^{-1}\left(\tan\frac{x_0}{2}\right)$$

$$=(2\pi-2x_0)+4\frac{x_0}{2}=2\pi$$

41. $(2P-3I)^4=-216P$

Take determinant on both sides we get

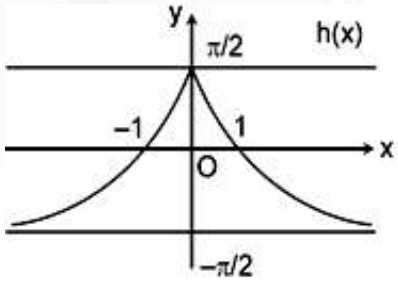
$|2P-3I|=\pm 16\times 27$ $\Rightarrow |2P-3PP^{-1}|=\pm 16\times 27$

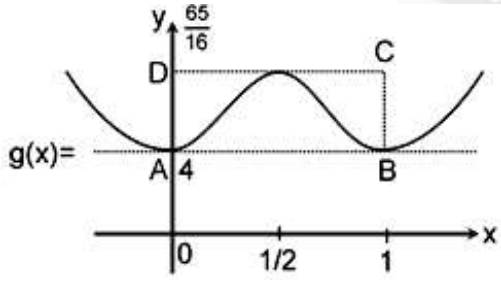
$|P||2I-3P^{-1}|=\pm 16\times 27$ $|2I-3P^{-1}|=\pm 27$

Now verify the options

42. $h(x)=\frac{\pi}{2}-\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ $h(x)=\begin{cases}\frac{\pi}{2}-2\tan^{-1}x & x\geq 0\\ \frac{\pi}{2}+2\tan^{-1}x & x\leq 0\end{cases}$

Clearly $a_1=0$



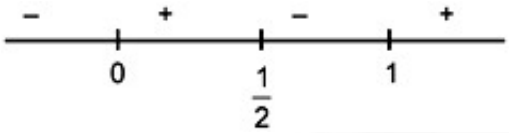


$g'(x)=4x^3-6x^2+2x$

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$\Rightarrow 2x\left(2x^2-3x+1\right)$ $=2x(2x-1)(x-1)$

$\Rightarrow a_1=0, a_2=\frac{1}{2}, a_3=1, a_4=3$



$g\left(\frac{1}{2}\right)=\frac{1}{16}-\frac{2}{8}+\frac{1}{4}+4=\frac{1}{16}+4$

$\therefore BC=\frac{1}{16}, AB=CD=1$ $\therefore$ Area of ABCD is $\frac{1}{16}\times 1=\frac{1}{16}$

Area $= 2\int_0^1 \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)dx = 2\int_0^1\left(\frac{\pi}{2}-2\tan^{-1}x\right)dx$

$=\pi-4\left[x\tan^{-1}x-\frac{1}{2}\ell og\left(1+x^2\right)\right]_0^1$

$=\pi-4\left[\left(\frac{\pi}{4}-\frac{1}{2}\ell og_e 2\right)\right]=\ell og_e 4$

43. We have $2[x]+64=3[x]-192\Rightarrow[x]=256$ $\Rightarrow x\in[256,257]$

Also, $\beta=\sin\frac{\pi}{18}\cdot\sin\frac{3\pi}{18}.....\sin\frac{9\pi}{18}.....\frac{17\pi}{18}$

$=\sin^2\frac{\pi}{18}\cdot\sin^2\frac{3\pi}{18}\cdot\sin^2\frac{5\pi}{18}\cdot\sin^2\frac{7\pi}{18}=\sin^2 10^\circ$

$\sin^2 30^\circ\cdot\sin^2 50^\circ\cdot\sin^2 70^\circ$

$\frac{1}{4}\left(\sin 10^\circ\cdot\sin 50^\circ\cdot\sin 70^\circ\right)=\frac{1}{4}\left(\frac{\sin 30}{4}\right)^2$ $\therefore\beta=\frac{1}{4}\left(\frac{1}{8}\right)^2=\frac{1}{256}$

44. $f(x)=x-2x^2$

$\therefore\int_0^{1/2}\left(x-2x^2\right)dx=\frac{1}{24}$ and area $=\left|\int_0^2\left(x-2x^2\right)dx\right|=\frac{10}{3}$

45. Given, $A+adj(A)=\alpha A^T$ ..... (1)

(A) If $\alpha=1, A+adj(A)=A^T$

$\Rightarrow adj(A)=-\left(A-A^T\right)$

$\Rightarrow\left|adj(A)\right|=\left|1-\left(A-A^T\right)\right|=0$

$\because A-A^T$ is skew symmetric.

(B) If $\alpha=1, A+adj(A)=A^T$

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$\Rightarrow 3A+adj(A)=A^T+2A$

So, $tr\left(3A+adj(A)\right)=tr\left(A^T+2A\right)\equiv tr(3A)$

= 9 (diagonal element are same in A^T and A)

(C) If $\alpha=-2, A+adj(A)=-2A^T$

$\Rightarrow 3A+Adj(A)=-2A^T+2A=2\left(A-A^T\right)$

$\Rightarrow \left|3A+Adj(A)\right|=0$

(D) If $\alpha=-2, A+adj(A)=-2A^T$

$\Rightarrow 4A+adj(A)=3A-2A^T$

So, $tr\left(4A+adj(A)\right)=tr\left(3A-2A^T\right)\equiv tr(A)=3$

46. $c_n=a_1+a_2+.....+a_n(d=2)$

$d_n=b_1+b_2+.....+b_n(d=2)$

$c_n=pa_n^2+qa_n+r$

$c_{n-1}=pa_{n-1}^2+qa_{n-1}+r$

From (1) and (2)

$c_n-c_{n-1}=p\left(a_n^2-a_{n-1}^2\right)+q(a_n-a_{n-1})$

$a_n=\left(p(a_n+a_{n-1})+q\right)(a_n-a_{n-1})$

Put $n=2\ a_2=d\left[p(a_2+a_1)+q\right]$

$a_3=d\left[p(a_3+a_2)+q\right]$

$a_3-a_2=d\left[p(a_3-a_1)\right]\Rightarrow p=\frac{1}{4}$

$c_n=pa_n^2=qa_n+r$

$a_1=c_1=pa_1^2+qa_1+r$

$c_2=a_1+a_2=pa_2^2+pa_2+r$

$a_2=a_1+2$

$2a_1+2=p(a_1+2)^2+q(a_1+2)+r$

$2a_1+2=\left(pa_1^2+qa_1+r\right)+4a_1p+4p+2q$

$2a_1+2=c_1+a_1+1+2q$

$2a_1+2=2a_1+1+2q \qquad q=\frac{1}{2}$

If $r=0, c_1=pa_1^2+qa_1$

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$c_1 = a_1 = \frac{1}{4}a_1^2 + \frac{1}{2}a_1$

$a_1^2 - 2a_1 = 0 \Rightarrow a_1 = 0, 2$

$d_1 = pb_1^2 + qb_1 \qquad (d_1 = b_1)$

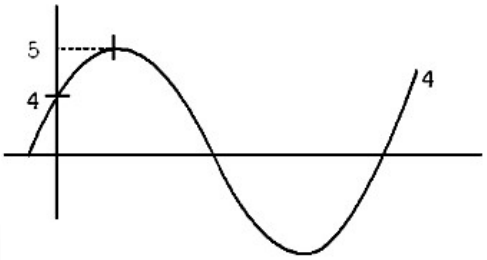
$b_1 = \frac{1}{4}b_1^2 + \frac{1}{2}b_1$

$b_1 = 0, 2$

$a_1 < b_1 \qquad (a_1 = 0 \,\&\, b_1 = 2)$

47. (A) Number of points of discontinuity

$\left[\frac{301}{3}\right] + \left[\frac{301}{5}\right] - 2\left[\frac{301}{15}\right] = 100 + 60 - 40 = 120$



Number of points of discontinuity is 19.

48. (A) $5\sec^{-1}\alpha + 10\sin^{-1}\beta = 10\pi \sec^{-1}\alpha = \pi$ and $\sin^{-1}\beta = \pi/2$

$\Rightarrow \alpha = -1; \beta = 1$, So $\tan^{-1}(-1) + \cos^{-1}(0) = -\frac{\pi}{4} + \frac{\pi}{2} = \frac{\pi}{4}$

(B) $\sin 2\pi x + \cos 2\pi x = \sin 2\pi x + \frac{1}{2} + \frac{\cos 2\pi x}{2}$

Min. value $= \frac{-\left(\sqrt{5}+1\right)}{2}$

(C) We have $x = \frac{\sqrt{3}\cos 20^\circ - 4\sin 20^\circ \cdot \cos 20^\circ}{\sin 20^\circ}$

$= \frac{2\sin 60^\circ \cdot \cos 20^\circ - 2\sin 40^\circ}{\sin 20^\circ} \qquad = \frac{2\sin 60^\circ \cdot \cos 20^\circ - 2\sin 40^\circ}{\sin 20^\circ} = 1$

(D) $f(x) = \sum_{n=1}^{4}\left[1 + \sin\frac{x}{n}\right] = \sum_{n=1}^{4}\left(1 + \left[\sin\frac{x}{n}\right]\right)$

$= 4 + [\sin x] + \left[\sin\frac{x}{2}\right] + \left[\sin\frac{x}{3}\right] + \left[\sin\frac{x}{4}\right]$

$\left[\sin\frac{x}{3}\right]$ and $\left[\sin\frac{x}{4}\right]$ is zero in $(0, p)$

Hence, $f(x) = 4 + [\sin x] + \left[\sin\frac{x}{2}\right]$

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Now range of $f(x)$ s $\{4, 5\}$

Hence sum if $4 + 5 = 9 \Rightarrow$ (D) is incorrect

49. $\frac{1}{\sqrt{k}} = \frac{2}{\sqrt{k}+\sqrt{k}} < \frac{2}{\sqrt{k}+\sqrt{k-1}} = 2\left(\sqrt{k}-\sqrt{k-1}\right)$

Putting k = 2, 3, ….. 10,000 and adding $\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+.....\frac{1}{\sqrt{10,000}} < 2\left(10,000-\sqrt{1}\right)$

$\Rightarrow A < 2\times 99 \Rightarrow A < 198$

Similarly $\frac{1}{\sqrt{k}} = \frac{2}{\sqrt{k}+\sqrt{k}} > \frac{2}{\sqrt{k+1}+\sqrt{k}} = 2\left(\sqrt{k+1}-\sqrt{k}\right)$

$\Rightarrow A > 2\left(\sqrt{10,000}-\sqrt{2}\right) > 197$

$\therefore [A] = 197 \qquad \therefore \left[\frac{[A]}{50}\right] = 3$

50. $n = \sum_{k=0}^{10} \binom{30+k}{k}\binom{20-k}{10-k}$

Using the standard binomial summation identify, $n = \binom{51}{10}$

$\left(\therefore usin g \sum_{r=0}^{k} \binom{n+r}{r}\binom{m-r}{k-r} = \binom{n+m+1}{k}\right)$

The number of trailing zeros equals the power of 5 in n.

$\upsilon_5(51!) = 12, \upsilon_5(10!) = 2, \upsilon_5 = (41!) = 9$

$\upsilon_5(n) = 12-(2+9) = 1$

51. If $x > 2$ then $x^3 - 3x > 4x - x = x > \sqrt{x+2}$

$\Rightarrow |x| \le 2$ take $x = 2\cos\theta$ for some $\theta \in [0, \pi]$

$\Rightarrow 2\cos 3\theta = \sqrt{2(1+\cos\theta)}$

$\Rightarrow 2\sin\frac{7\theta}{4}\cdot\sin\frac{5\theta}{4} = 0$

$\Rightarrow \theta = 0, \frac{4\pi}{7}, \frac{4\pi}{5}$

$\therefore x = 2, 2\cos\frac{4\pi}{7}, -\frac{1}{2}\left(1+\sqrt{5}\right)$

52. $E = \sum_{i=1}^{n} \frac{n}{\frac{1}{a_{i+1}^{2/3}}+\frac{1}{a_i^{2/3}}+\frac{1}{(a_i a_{i+1})^{1/3}}}$

Let $\frac{1}{a_{i+1}^{1/3}} = A \qquad \frac{1}{a_i^{1/3}} = B$

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$$E=\sum_{i=1}^{n}\frac{n}{A^2+B^2+AB}--=\sum_{i=1}^{n}\frac{n(A-B)}{A^3-B^3}$$

$$E=n\sum_{i=1}^{n}\frac{\left(\frac{1}{a_{i+1}^{1/3}}-\frac{1}{a_i^{1/3}}\right)}{d}$$

$$\frac{1}{a_{i+1}}-\frac{1}{a_i}=d$$

$$E=\frac{n}{d}\sum_{i=1}^{n}\left(\frac{1}{a_{i+1}^{1/3}}-\frac{1}{a_i^{1/3}}\right)$$

$$\Rightarrow E=\frac{n}{d}\left[\frac{1}{2}-\frac{1}{1}\right]=-\frac{n}{2d} \text{ also } \frac{1}{8}=\frac{1}{a_1}+nd\Rightarrow d=-\frac{7}{8n}$$

$$E=\frac{n\cdot 8n}{-7}\left(\frac{1}{2}-1\right)=\frac{4n^2}{7}$$

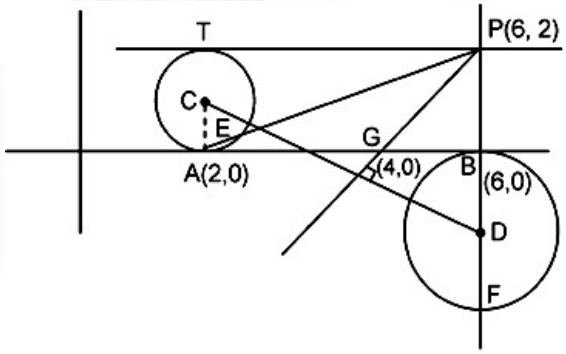
53. $a^2=13\lambda+r_1 \qquad r_1\in\{0,\pm1,\pm2,.....,\pm6\}$

$$b^2=13\lambda+r_2$$

$$c^2=13\lambda+r_3$$

$$P=\frac{fav}{Total}=\frac{1+4\times3!+4\times3!+8\times3!+4\times3!+8+3!}{13^3}$$

$$P=\frac{169}{13^3}=\frac{1}{13}$$



54.

Clearly P lies on radical axis

$$\Rightarrow PE\cdot PA=PB\cdot PF$$

$$\Rightarrow\frac{PE}{PF}=\frac{PB}{PA}=\frac{2}{2\sqrt{5}}=\frac{1}{\sqrt{5}}$$

$$PE\cdot PA=PT^2$$

$$\Rightarrow PE=\frac{16}{2\sqrt{5}}=\frac{8}{\sqrt{5}}$$

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